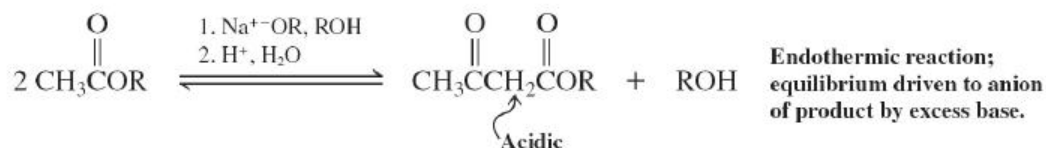


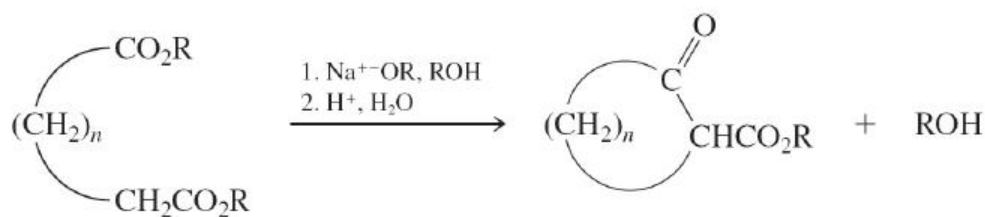
New Reactions

Synthesis of $\beta\beta$ -Dicarbonyl Compounds

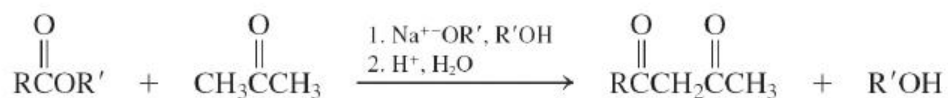
1. Claisen Condensation ([Section 23-1](#))



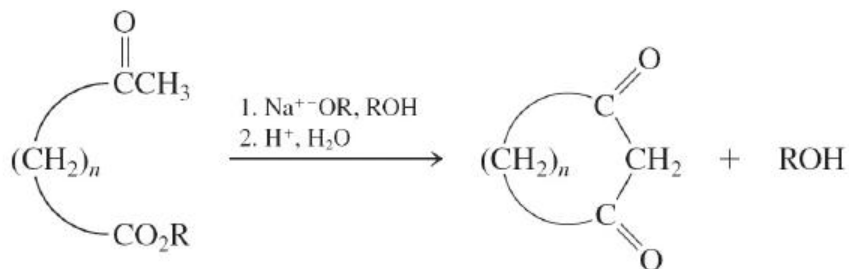
2. Dieckmann Condensation ([Section 23-1](#))



3. $\beta\beta$ -Diketone Synthesis ([Section 23-1](#))

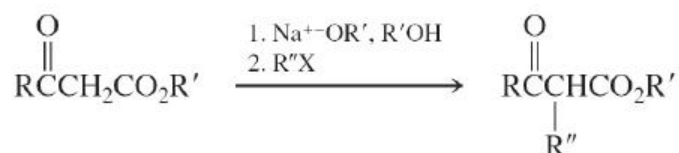


Intramolecular

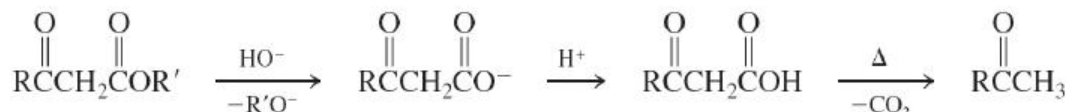


3-Ketoesters as Synthetic Building Blocks

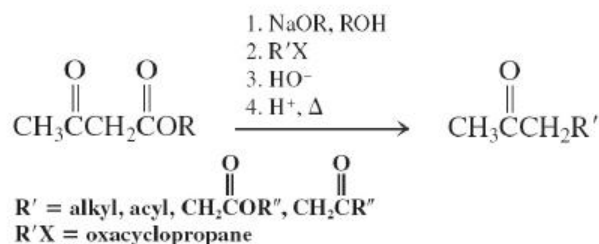
4. Enolate Alkylation ([Section 23-2](#))



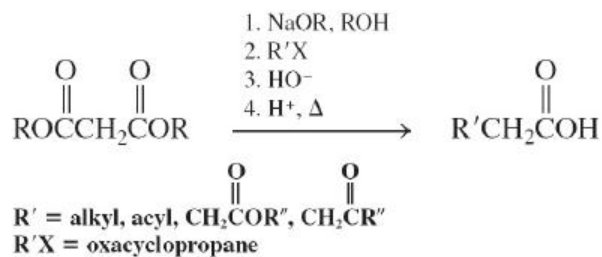
5. 3-Ketoacid Decarboxylation ([Section 23-2](#))



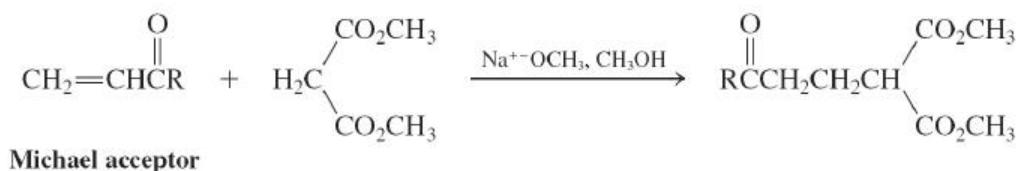
6. Acetoacetic Ester Synthesis of Methyl Ketones ([Section 23-2](#))



7. Malonic Ester Synthesis of Carboxylic Acids ([Section 23-2](#))

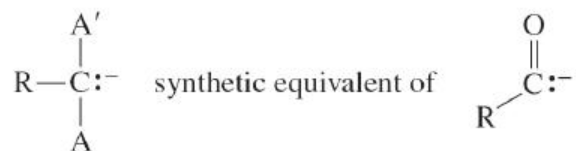


8. Michael Addition ([Section 23-3](#))

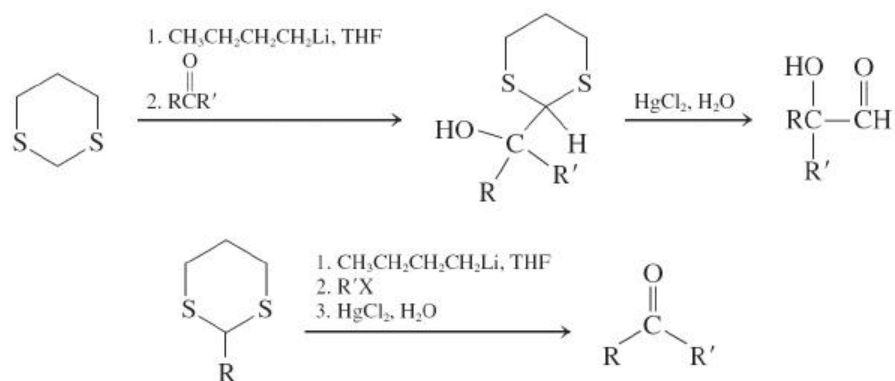


Acyl Anion Equivalents

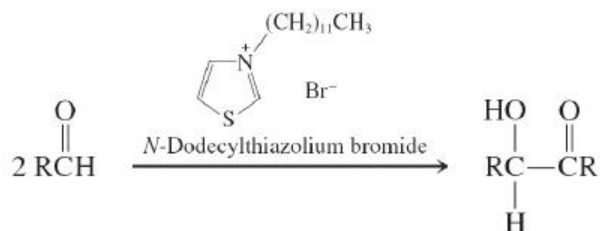
9. 1,3-Dithiacyclohexane (1,3-Dithiane) Anions as Acyl Anion Equivalents ([Section 23-4](#))



A = electron-withdrawing, conjugating, or polarizable group



10. Thiazolium Salts in Aldehyde Coupling ([Section 23-4](#))



Important Concepts

1. The **Claisen condensation** is driven by the stoichiometric generation of a stable $\beta\beta$ -**dicarbonyl anion** in the presence of excess base.
2. $\beta\beta$ -**Dicarbonyl compounds** contain acidic hydrogens at the carbon between the two carbonyl groups because of the inductive electron-withdrawing effect of the two neighboring carbonyl functions and because the anions resulting from deprotonation are resonance stabilized.
3. Although **mixed Claisen condensations** between esters are usually not selective, they can be so with certain substrates (nonenolizable esters, intramolecular versions, ketones).
4. 3-Ketoacids are unstable; they **decarboxylate** in a concerted process through an aromatic transition state. This property, in conjunction with the nucleophilic reactivity of 3-ketoester anions, allows the synthesis of substituted ketones and acids.
5. Because **acyl anions** are not directly available by deprotonation of aldehydes, they have to be made as masked reactive intermediates or stoichiometric reagents by transformations of functional groups.